Sales and Profit Analysis Using SQL and Tableau

A complete business intelligence workflow transforming raw transactional data into actionable insights

Project Overview

Raw Data

150,281 transactions across 5 tables from 2017-2020

Data Processing

SQL-based cleaning, transformation, and star schema modeling

Visual Analysis

Interactive Tableau dashboards for revenue and profit insights

This project demonstrates an end-to-end analytics workflow, from data backup and cleaning through SQL analysis to interactive dashboard creation. The work is organized into two analytical flows: revenue analysis and profit analysis, each presented independently.

Dataset Structure

Star Schema Design

One fact table (transactions) connected to four dimension tables (customers, products, markets, date). This structure enables efficient analysis across multiple business dimensions.

Key Statistics

- 150,281 transaction records
- 38 customers, 279 products
- 17 markets, 1,126 dates
- 4 years of data (2017-2020)

Revenue Dashboard Highlights

986.62M

Total Revenue

Combined sales across all markets and years

2,431.54K

Sales Quantity

Units sold throughout the period

520.77M

Delhi NCR Revenue

Top market contribution

Profit Dashboard Insights

24.66M	4.88%	-20.78%
Total Profit	Surat Margin	Bangalore Margin
Net earnings after costs	Highest profit margin	Negative profitability

The profit dashboard reveals a critical insight: revenue and profit margin do not follow the same pattern. High sales volume doesn't guarantee profitability.

Key Business Insights

Market Concentration

Delhi NCR dominates with 520.77M revenue and 999.90K units sold, followed by Mumbai (150.81M) and Ahmedabad (172.31M)

Customer Dependency

Top customer (ElectricalsStores) contributes 413.91M, creating concentration risk that requires diversification

Channel Mix

Brick & Mortar accounts for 75.60% of business, while E-Commerce represents 24.40% growth opportunity

Profit Variability

Surat shows 4.88% margin while Bangalore has -25.78%, revealing significant profitability differences across markets

Business Recommendations

01

Focus on High-Revenue Markets

Maintain strong relationships and supply chain efficiency in Delhi NCR, Mumbai, and Ahmedabad

02

Improve Low-Margin Markets

Review pricing strategy, transportation costs, and operational expenses in markets like Bangalore

03

Expand High-Margin Markets

Consider expanding operations in Surat, Patna, and Bhubaneswar where profit margins are strongest

04

Diversify Customer Base

Reduce dependency on top customers while maintaining strong relationships with key accounts

05

Monitor Profit with Revenue

Track profit margin alongside revenue before making business decisions to ensure sustainable growth

Tools and Methodology

MySQL Workbench

Data import, backup creation, cleaning, star schema building, and SQL analysis queries

Tableau Public

Interactive dashboard creation and visualization of revenue and profit analysis results

Star Schema

Fact table with transaction measures connected to dimension tables with descriptive attributes

Future Enhancements

Predictive Analytics

Time series forecasting for sales and profit trends

Customer Segmentation

RFM analysis for customer lifetime value

Live Dashboard

Automated ETL pipeline with automatic data refresh

Current limitations include reliance on historical data (2017-2020), descriptive analytics only, and static dashboards requiring manual refresh. Future scope includes forecasting, advanced segmentation, and live data connections.

Conclusion

This project demonstrates how raw transactional data can be transformed into useful business information through structured SQL processing and Tableau dashboarding.

Complete Workflow

From data understanding and backup through cleaning, modeling, and analysis to interactive dashboards

Business Value

Not only shows sales numbers but reveals profitability patterns across customers, markets, products, and time

Actionable Insights

Identifies market opportunities, customer risks, and profitability challenges for better decision-making